

K GOUTAM

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student passionate about Defence Technologies, working at the intersection of embedded hardware, network security, and Machine Learning to build intelligent, data-driven defence solutions. Demonstrated leadership through hackathon wins, class representation, and national-level event organisation.

EDUCATION

Bachelor of Engineering - Electronics & Communication Engineering
Vidyavardhaka College of Engineering, Mysuru | CGPA (till 6th sem): 7.73
Class XII - AISSCE (PCM + Computer Science)
Sri Sankara Vidyalaya, Bhilai | Score: 358/500 (71.6%)

TECHNICAL SKILLS

Languages: Python, SQL

Frameworks & Tools: Flask, scikit-learn, Pandas, NumPy, Data Visualisation Tools, Raspberry Pi, Mosquitto (MQTT), iptables, Linux

Domains: Machine Learning, IoT & Network Security, Embedded Systems, Web Development

Soft Skills: Team Leadership, Problem Solving, Event Management, Cross-functional Collaboration

PROJECTS

MUD-Based IoT Security System - RFC 8520 Implementation

Skills: Python, Raspberry Pi, Linux, iptables, MQTT (Mosquitto), Network Security, JSON, Flask

- Implemented the IETF RFC 8520 Manufacturer Usage Description (MUD) standard on Raspberry Pi hardware to automatically restrict IoT devices to their manufacturer-declared network behaviour.
- Built a Flask-based MUD file server and a MUD manager that fetches, verifies, and parses MUD policy files, translating declared ACLs into live iptables firewall rules on the gateway.
- Added HMAC-SHA256 MUD file signature verification and CSV audit logging of every policy decision; validated enforcement via rogue-device simulation with 8/8 correct allow/block outcomes.

Competitor Review Intelligence Platform

Skills: Python, Web Scraping, NLP, Generative AI, Data Visualization, REST APIs

- Built an AI-powered competitor intelligence platform that aggregates and analyses customer reviews from Zepto, BigBasket, and Swiggy Instamart.
- Delivers actionable insights on customer sentiment, product pain points, and feature-level gaps to support strategic business decisions.

Skin Cancer Detector using Machine Learning

Skills: Python, Deep Learning, Kaggle Dataset, Image Processing

- Built a deep learning model that classifies dermoscopic images to detect skin cancer with high accuracy.

Symbiot 2026 - National Hackathon Official Website

Skills: Web Development, Full-Stack, Render, Responsive Design

- Designed and deployed the official website for Symbiot 2026, a 24-hour national hackathon at VVCE Mysuru, featuring problem-statement management, team shortlisting, live results, domain-wise filtering, and a mobile-optimised UI.

ACHIEVEMENTS & AWARDS

- Class Representative, ECE Department, VVCE (July 2025 - Present) - Elected class liaison between students and faculty, coordinating schedules, feedback, and departmental communication.
- Runner-Up, 'Idea to Enterprise' Competition (ASPERA VVCE) - Presented a Modular Laptops concept focused on user upgradeability, customisation, and e-waste reduction in a 12-hour business innovation sprint.
- Best Achiever in Co-Curricular Activities, 4th Semester, VVCE - Recognised for academic contributions and participation beyond the classroom.
- Letter of Appreciation for building the official website of Symbiot 2026, organised by the ECE Department.

CERTIFICATIONS

- Supervised Machine Learning: Regression & Classification | Advanced Learning Algorithms - Stanford Online (Coursera)
- Networking Basics - Cisco Networking Academy | Introduction to Databases - Meta (Coursera)
- Python for Everybody - Coursera (University of Michigan) | MATLAB Onramp - MathWorks
- Aerial Robotics - University of Pennsylvania (Coursera) | Introduction to Microprocessors - ARM
- Data Science with Generative AI - PW Skills | Issued Sep 2026

VOLUNTEERING & LEADERSHIP

Roles: Core Member, RoboRise Robotics Club (May 2025-Present) | Member, IoTcrew, VVCE (Aug 2024-Present) | Volunteer, Youth for Seva

- Organiser, Symbiot 2026 - managed logistics, judging, and live operations; participated in 5+ hackathons and technical workshops.